



RIVER VALLEY HIGH SCHOOL

YEAR 6 PRELIMINARY EXAMINATION

CANDIDATE
NAME

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CENTRE
NUMBER

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INDEX
NUMBER

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CLASS

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H2 BIOLOGY

9648/02

Paper 2 Core Paper

20 September 2013

2 hours

Additional Materials: Answer Paper

READ THESE INSTRUCTIONS FIRST

Do not open this booklet until you are told to do so.

Write your Centre number, index number and name on all the work you hand in.

Write in dark blue or black pen on both sides of the paper.

You may use a soft pencil for any diagrams, graphs or rough working.

Do not use staples, paper clips, highlighters, glue or correction fluid.

Section A

Answer **all** questions.

Section B

Answer any **one** question.

At the end of the examination, fasten all ruled paper securely.

Hand in the Question Paper and answers to Section B separately.

The number of marks is given in brackets [] at the end of each question or part question.

For Examiner's Use	
Section A	
1	/10
2	/ 12
3	/ 12
4	/ 12
5	/ 12
6	/ 10
7	/ 12
Section B	
8 or 9*	/ 20
Total	

This Question Paper consists of **25** printed pages.

Section A (80 marks)

Answer **all** the questions in this section.

1. **Fig. 1.1** is a diagram of an electron micrograph of part of a plant cell.

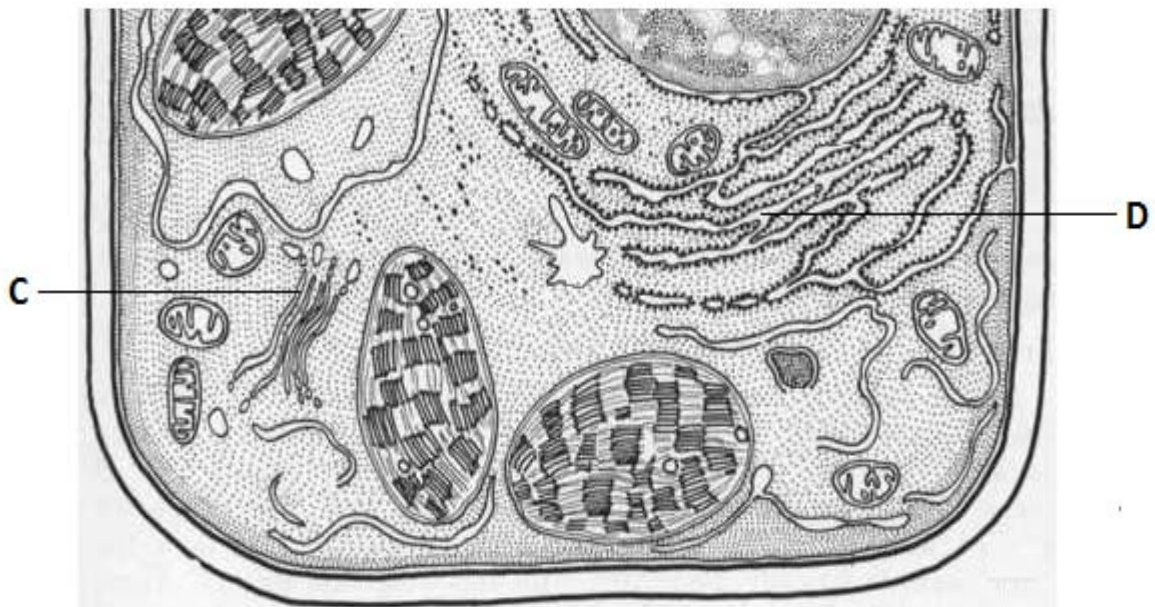


Fig. 1.1

- (a) Using lines, indicated precisely in **Fig. 1.1**, a possible location where each of the following molecules can be found:

A: chlorophyll

B: cellulose

[2]

- (b) (i) Identify the organelle labelled **C** and state a visible feature that enabled your identification.

[2]

1. (b) (ii) Explain how the functions of the two organelles, **C** and **D**, are linked. [2]

Fig. 1.2 shows a magnified image of organelle **D**.

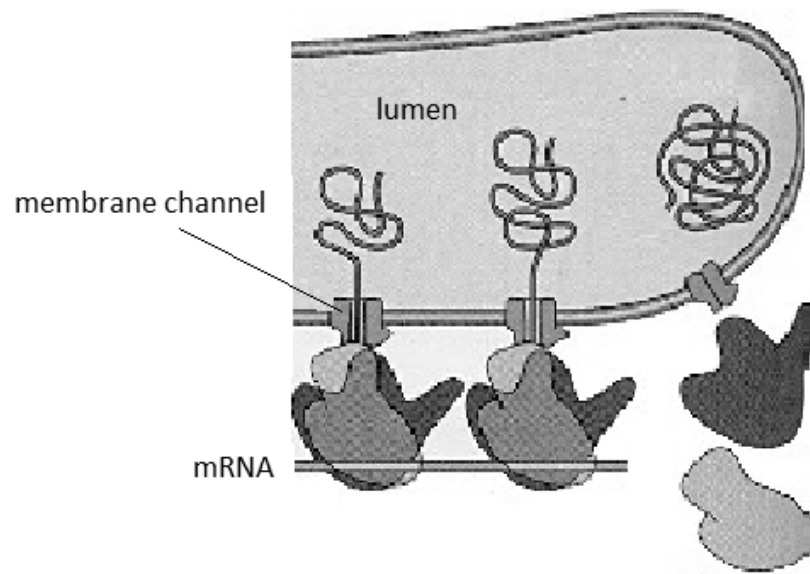


Fig. 1.2

- (c) (i) Describe the function of the protein that forms the membrane channel. [3]

1. (c) (ii) Suggest how prokaryotes, which have no membrane-bound organelles, are able to secrete proteins. [1]

[Total: 10]

2. Yeast haploid cells secrete pheromones to signal mating, and respond by growing a mating projection towards a potential mate. Upon contact of the two partner cells, these fuse to form a diploid zygote.

Fig. 2.1 shows the pheromone-signaling pathway mediated by yeast G-protein coupled receptor, Ste2. The activation of the pathway induces the expression of *FUS1* gene which is required for yeast mating.

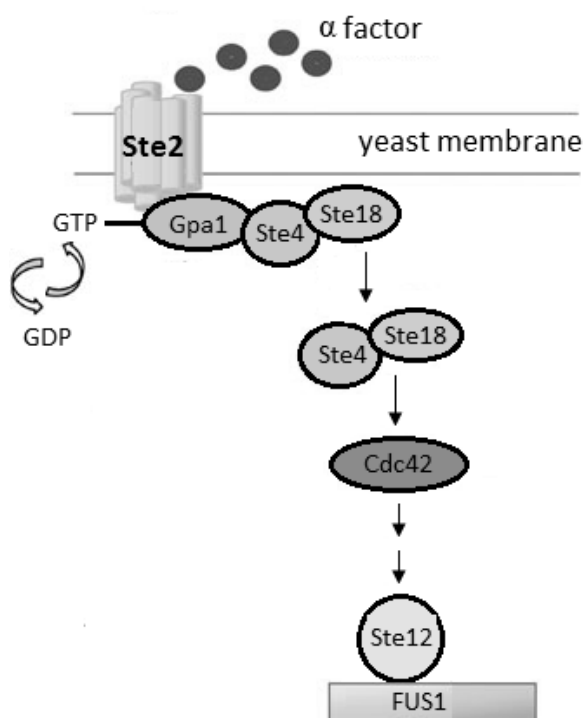


Fig. 2.1

- (a) With reference to **Fig. 2.1**, describe how α factor triggers the activation of Cdc42.

[4]

2. (b) Suggest why α factor cannot enter the yeast cell. [2]

Fig. 2.2 shows the Ste2 receptor on another yeast cell membrane.

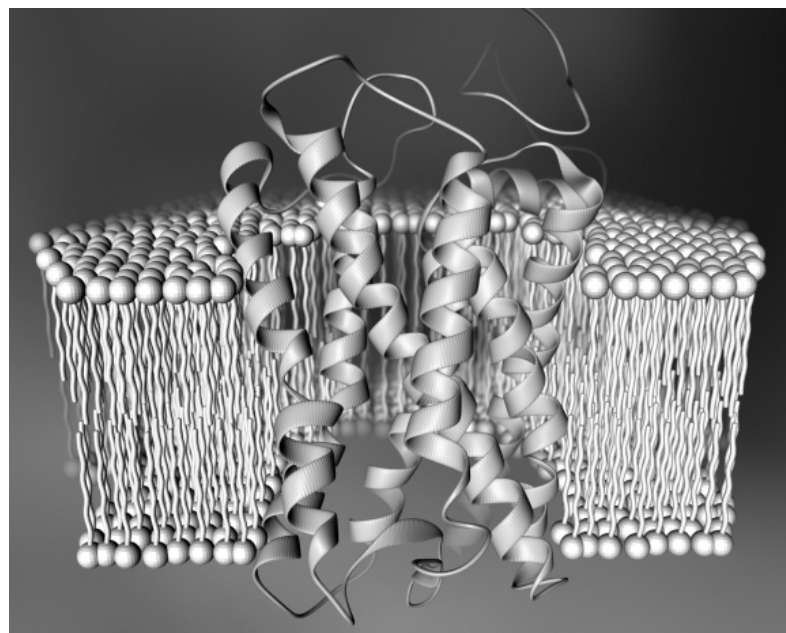


Fig. 2.2

- (c) Describe the structure of Ste2 receptor. [3]

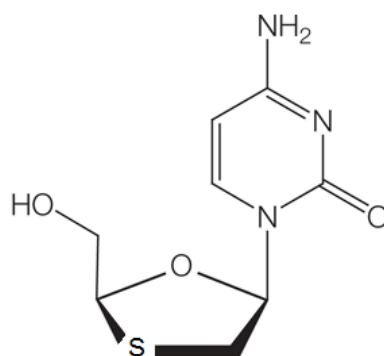
2. (d) Explain how Ste2 receptor remains embedded in the yeast cell membrane. [3]

[Total: 12]

3.

PART I

Lamivudine is an antiviral medication that prevents Human Immunodeficiency Virus (HIV) from multiplying in the body. The structure of lamivudine, an analogue of viral nucleotide, is shown in **Fig. 3.1**.



Lamivudine

Fig. 3.1

- (a) Explain how lamivudine prevents HIV from multiplying in the body. [2]

3. Many drugs have been designed to make it possible for AIDS patients to improve their overall health and extend their lives. However, most of these drugs have low efficacy due to high variance in the HIV genome. **Fig. 3.2** shows the partial cDNA sequence of glycoprotein 120 (gp120) on two different strains of HIV found in a host cell. The difference in the two DNA sequences is indicated via a box.

HIV Strain 1

```
61 ctgaaaatct cgaaaacaat gccaaaacca taatagtgc ccttaataaa tctgtagaaa
121 tcaattgtac cagaccctcc aacaattacaa gaacaagtat aactatagga ccaggacaag
181 tatggtatag aacaggagac ataataggaa atataagaaa agcatattgt gagattaatg
241 gaacaaaatg gaataaagtt taaaacagg taactgaaaa attgaaagag cactttaata
301 ataagacaat aatctttcaa ccac
```

HIV Strain 2

```
61 ctgaaaatct cgaaaacaat gccaaaacca taatagtgc ccttaataaa tctgtagaaa
121 tcaattgtac cagaccctcc aacactacaa gaacaagtat aactatagga ccaggacaag
181 tatggtatag aacaggagac ataataggaa atataagaaa agcatattgt gagattaatg
241 gaacaaaatg gaataaagtt taaaacagg taactgaaaa attgaaagag cactttaata
301 ataagacaat aatctttcaa ccac
```

Fig. 3.2

- (b) Explain how the genomic variation shown in **Fig. 3.2** affects the structure of gp120 on the two strains of HIV.

[3]

3. (c) Describe how HIV enters host cell. [3]

PART II

Fig. 3.3 shows the structure of an isolated virus.

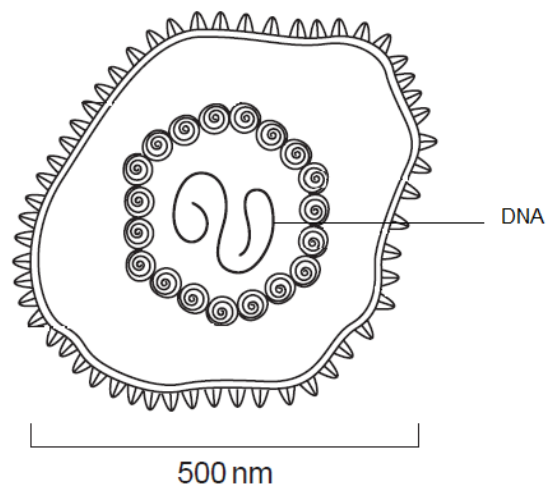


Fig. 3.3

- (d) State two **visible** structural similarities between influenza virus and the virus shown in **Fig. 3.3**. [2]

3. (e) Describe how the genome of influenza virus differs from that of the virus shown in **Fig. 3.3**. [2]

[Total: 12]

4.

PART I

Fig. 4.1 is an electron micrograph of chromatin maintained at the 30 nm higher-order fiber.



Fig. 4.1

(a) Most of the chromatin is seen to be in the form of a fiber with a diameter of about 30 nm.

(i) Describe how DNA is packed into the 30 nm chromatin fibre. [2]

(ii) Explain why the genome is packaged this way. [2]

4. Processes such as transcription and replication require the two strands of DNA to come apart temporarily, thus allowing access of polymerases to the DNA template. However, the folding of chromatin into 30 nm fibers poses barriers to the enzymes that unwind and copy DNA.

(b) Explain how the histones are removed transiently to permit replication or transcription to proceed. [3]

PART II

Fig. 4.2 and **Fig. 4.3** are diagrams showing DNA replication and transcription, respectively.

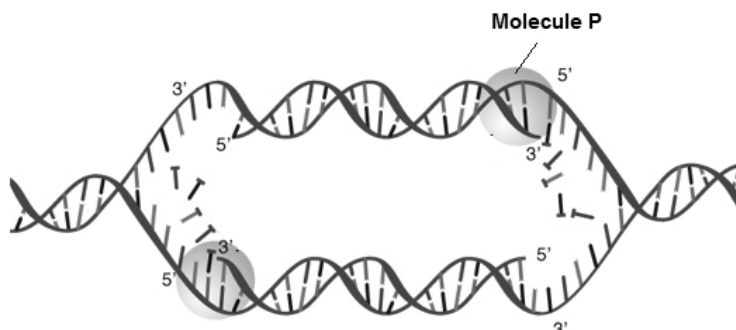


Fig. 4.2

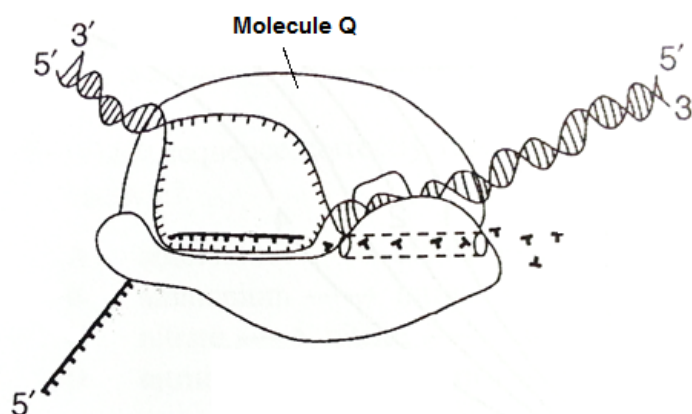


Fig. 4.3

4. (c) Explain why RNA primer is required for the synthesis of new strand in DNA replication. [1]

- (d) Describe **two** ways in which the functions of molecules **P** and **Q** are

- (i) similar to each other; [2]

- (ii) different from each other. [2]

[Total: 12]

5. In a particular variety of tomato plant, the allele for red fruit colour (**A**) is dominant over the allele for orange fruit (**a**) and the allele for green base when ripe (**B**) is dominant over the allele for no green base when ripe (**b**).

Two students, Eugene and Cornelius crossed plant with red fruit and green base when ripe with pure bred plant with orange fruit and no green bases when ripe. The phenotypes of 50 offspring of each of Eugene's and Cornelius's crosses were recorded and are shown in **Table 5.1**.

Table 5.1

	Phenotypes of offspring of test crosses			
	Red fruit with green base	Red fruit with no green base	Orange fruit with green base	Orange fruit with no green base
Eugene's cross	23	4	3	20
Cornelius's cross	3	21	23	3

- (a) Describe briefly how a plant breeder ensures that the offspring produced are **only** from the desired cross. [2]

- (b) Using the results from Eugene's cross,

- (i) Carry out a χ^2 test using the formula below. [1]

$$\chi^2 = \sum \frac{(O - E)^2}{E}$$

5. (b) (ii) With reference to the table of probabilities provided, explain how the value for the χ^2 test supports the hypothesis that the two genes are independently segregating.

[2]

degrees of freedom	probability, p				
	0.10	0.05	0.02	0.01	0.001
1	2.71	3.84	5.41	6.64	10.83
2	4.61	5.99	7.82	9.21	13.82
3	6.25	7.82	9.84	11.35	16.27
4	7.78	9.49	11.67	13.28	18.47

- (c) Using the symbols given, state the genotypes of the tomato plants in Eugene's and Cornelius's crosses.

[2]

Eugene's cross:

Cornelius's cross:

5. (d) Draw a genetic diagram to explain the results of Cornelius's cross.

[5]

[Total: 12]

6.

PART I

Fig. 6.1 is a diagram of an electron micrograph of part of a chloroplast showing thylakoid membranes.

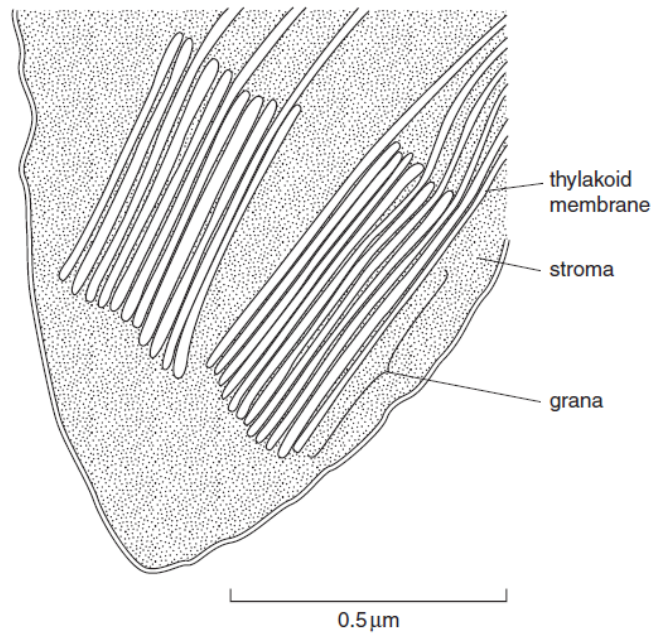


Fig. 6.1

(a) Describe the role of thylakoid membrane in photosynthesis. [4]

6.

PART II

Arabidopsis thaliana is a small flowering plant native to Asia. A mutation in the gene coding for NADP oxidase results in plants with short root hairs. NADP oxidase is an enzyme that converts NADPH to NADP⁺.

Fig. 6.2 shows the root hairs in the two variant of *A. thaliana*.

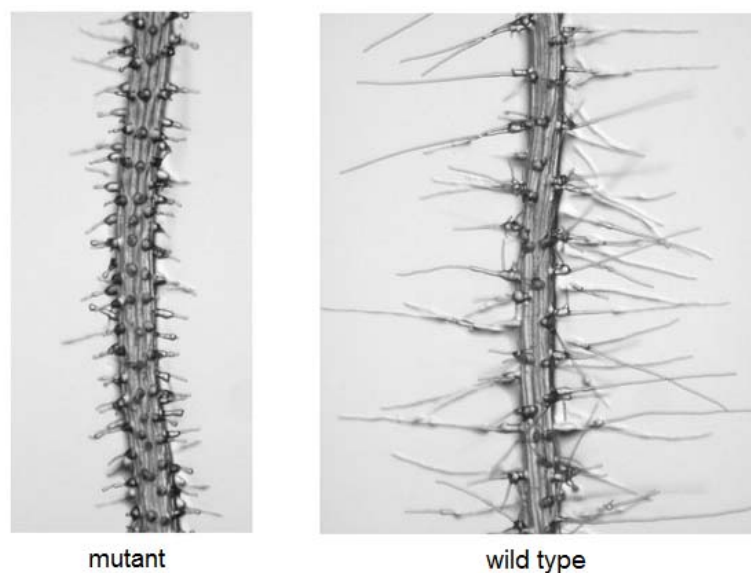


Fig. 6.2

(b) Explain the role of NADPH in photosynthesis.

[2]

6. In a separate experiment, the length of root hairs in wild type and mutant *A. thaliana* were measured at intervals, together with the activity of NADP oxidase in the tips of the root hairs. The results are shown in **Fig. 6.3**.

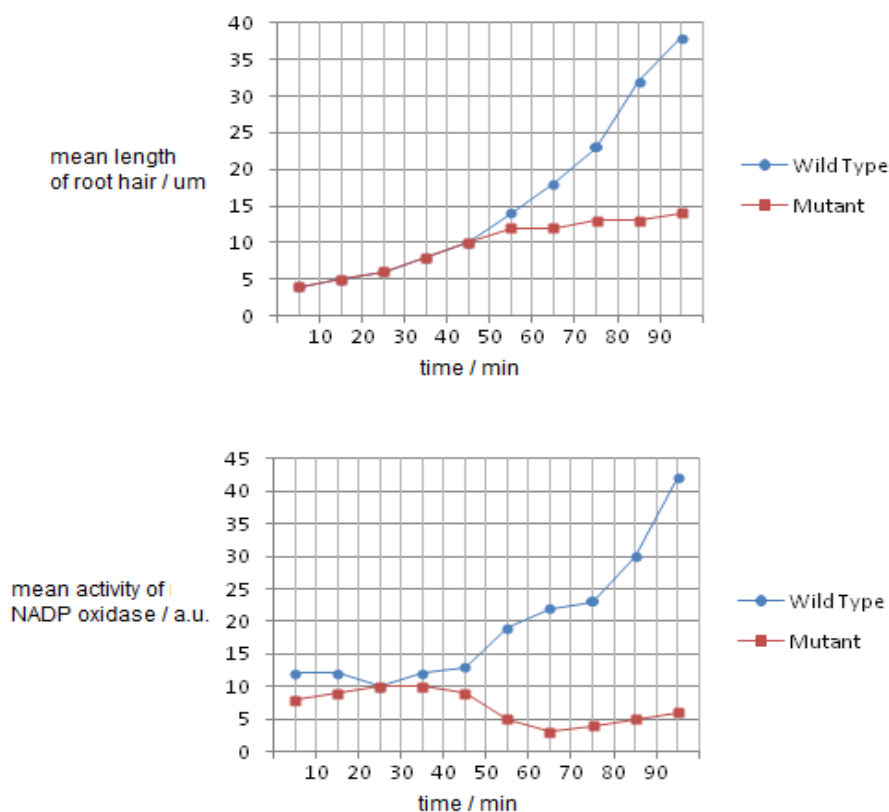


Fig. 6.3

- (c) With reference to **Fig. 6.3**, and your knowledge of photosynthesis, describe and explain the difference in the growth of root hairs in the two types of *A. thaliana*.

[4]

[Total: 10]

7. **Fig. 7.1** shows the arrangement of bones in the pentadactyl forelimbs of four vertebrates.

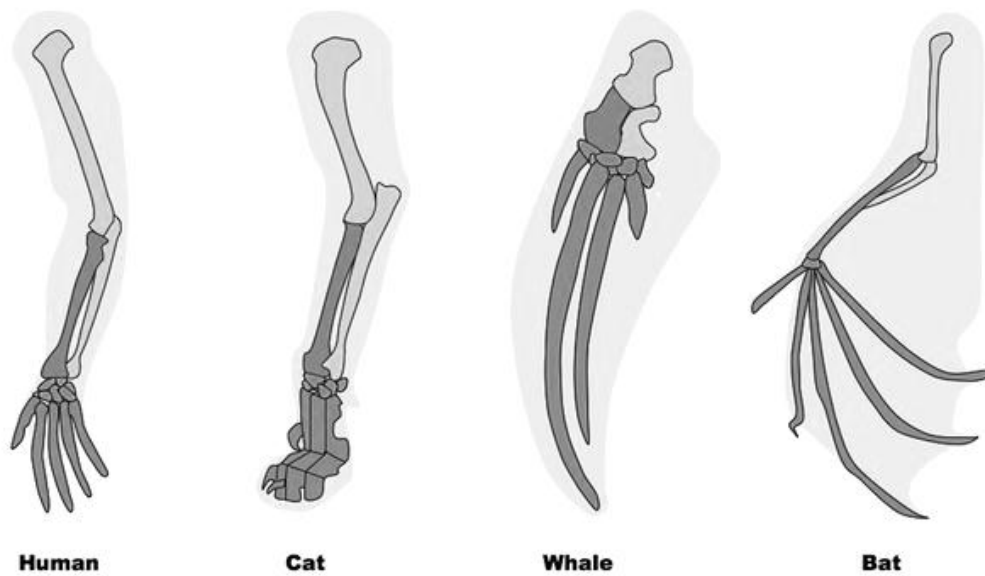


Fig. 7.1

- (a) Explain how the relationship between the structures in **Fig. 7.1** provides evidence to support the theory of evolution. [2]

- (b) Explain how fossil evidence can also be used as evidence to support the idea that evolution has taken place? [2]

7. (c) Suggest how natural selection could have brought about the evolution of the skeleton of a bat's wing.

[3]

The pipistrelle is the most common species of bat in Europe. **Fig. 7.2** shows a map of Canary Islands, a Spanish archipelago located just off the northwest coast of mainland Africa. El Hierro is the most recent Canary Island, having appeared approximately 1.2 million years ago.

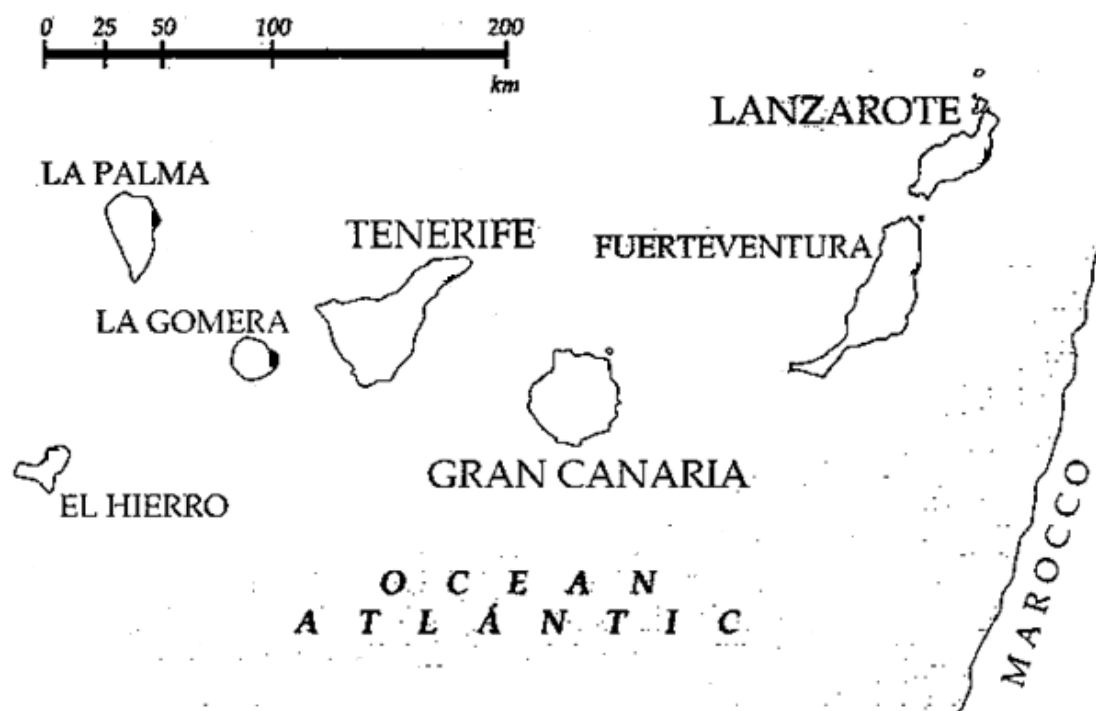


Fig. 7.2

7. It was originally thought that all pipistrelles belonged to the same species. However, mtDNA reveals that there were two species: *Pipistrellus maderensis* and *Pipistrellus kuhlii*. The data for both species are provided in **Table 7.1**.

Table 7.1

Species	Mean body mass (g)	Mean wingspan (cm)	Colour
<i>P. maderensis</i>	5.7	30	dark brown fur with light wing markings
<i>P. kuhlii</i>	5.5	30	light brown to reddish brown with light wing markings

- (d) Using the data in **Table 7.1**, suggest why pipistrelles were originally classified as one species. [1]

Evolution of the two bat species, *Pipistrellus maderensis* and *Pipistrellus kuhlii*, was studied by comparing ~1 kbp of mtDNA (from *cytochrome b* and 16S rRNA genes) from the bats in different islands.

Sequence divergence of *P. maderensis* is the lowest between La Gomera and El Hierro as compared to between La Gomera and the remaining islands (i.e. Tenerife and La Palma).

- (e) Using the above information and **Fig. 7.2**,

- (i) explain why the sequence divergence of *P. maderensis* is the lowest between La Gomera and El Hierro. [3]

7. (e) (ii) suggest why no gene flow occurs between the *P. maderensis* on La Gomera and El Hierro. [1]

[Total: 12]

Section B (20 marks)

Answer **one** question.

Write your answers on the separate answer paper provided.
Your answers should be illustrated by large, clearly labelled diagrams, where appropriate.
Your answers must be in continuous prose, where appropriate.
Your answers must be set out in sections **(a)**, **(b)** etc., as indicated in the question.

A **NIL** return is necessary if you have not attempted this section.

8. (a) Describe the structure of the cell membrane and relate it to its properties. [7]
(b) Explain the significance of mitosis in growth, repair and asexual reproduction. [4]
(c) Outline the behaviour of chromosomes during meiosis I and explain the significance of each phase in leading to genetic variation. [9]

[Total: 20]

9. (a) Discuss, using named examples, how dual control of gene expression is achieved in *Escherichia coli*. [8]
(b) Explain the significance of operons in bacteria. [4]
(c) Explain how the structure of neurone facilitates impulse transmission. [8]

[Total: 20]